

Student 9 **Gudi Yashwanth** 192524275RD

10 / 10 submitted

Q1. Selection Sort

Score: 10.00 TC: 3/3 passed

CODE

```
nums = list(map(int, input().split()))

n = len(nums)
for i in range(n):
    min_index = i
    for j in range(i + 1, n):
        if nums[j] < nums[min_index]:
            min_index = j
    nums[i], nums[min_index] = nums[min_index], nums[i]

print(nums)
```

INPUT

23 78 45 8 32 56

OUTPUT

[8, 23, 32, 45, 56, 78]

Q2. Insertion Sort

Score: 10.00 TC: 3/3 passed

CODE

```
def insertion_sort(arr):
    for i in range(1, len(arr)):
        key = arr[i]
        j = i - 1
        while j ≥ 0 and arr[j] > key:
            arr[j + 1] = arr[j]
            j -= 1
        arr[j + 1] = key
    return arr

# Reading input
arr = list(map(int, input().split()))
result = insertion_sort(arr)
print(result)
```

INPUT

8 5 7 1 9 3

OUTPUT

[1, 3, 5, 7, 8, 9]

Q3. Merge Sort

Score: 10.00 TC: 3/3 passed

CODE

```
def merge_sort(arr):
    if len(arr) ≤ 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])

    return merge(left, right)

def merge(left, right):
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i] ≤ right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])

    return result

# Read input
arr = list(map(int, input().split()))

# Sort and print
sorted_arr = merge_sort(arr)
print(sorted_arr)
```

INPUT

38 27 43 3 9 82 10

OUTPUT

[3, 9, 10, 27, 38, 43, 82]

Q4. Diagonal Matrix

Score: 6.67 TC: 2/3 passed

CODE

```
s = input().strip()

s = s.replace("[", "").replace("]", "")

numbers = list(map(int, s.split(",")))

n = int(len(numbers) ** 0.5)

diagonal = True

for i in range(n):
    for j in range(n):
        if i != j and numbers[i * n + j] != 0:
            diagonal = False

if diagonal:
    print("Diagonal Matrix")
else:
    print("Not a Diagonal Matrix")
```

INPUT

[[1,0,0],[0,5,0],[0,0,8]]

OUTPUT

Diagonal Matrix

Q5. Row and Column Sums

Score: 6.67 TC: 2/3 passed

CODE

```
s = input().strip().replace(" ", "")

s = s[1:-1]

parts = s.split(",")

matrix = []

for part in parts:
    part = part.replace("[", "").replace("]", "")
    matrix.append([int(x) for x in part.split(",")])

rows = []
cols = []

for i in range(len(matrix)):
    rows.append(sum(matrix[i]))

for j in range(len(matrix[0])):
    total = 0
    for i in range(len(matrix)):
        total += matrix[i][j]
    cols.append(total)

print("Row=", rows)
print("Col=", cols)
```

INPUT

[[1,2],[3,4]]

OUTPUT

Row= [3, 7]
Col= [4, 6]

Q6. Maximum Subarray Sum

Score: 6.67 TC: 2/3 passed

CODE

```
# Get the input list of integers
nums = list(map(int, input().split()))

# Kadane's Algorithm - Simplified
current_max = max_so_far = nums[0]

for x in nums[1:]:
    current_max = max(x, current_max + x)
    max_so_far = max(max_so_far, current_max)

print(max_so_far)
```

INPUT

-1 -2 -3

OUTPUT

-1

Q7. Common Elements Among Three Lists

Score: 10.00 TC: 3/3 passed

CODE

```
a, b, c = input().strip().split(';')

list1 = [int(x) for x in a.split()]
list2 = [int(x) for x in b.split()]
list3 = [int(x) for x in c.split()]

common = sorted(set(list1) & set(list2) & set(list3))
print(common)
```

INPUT

1 2 3;2 3 4;2 5

OUTPUT

[2]

Q8. Matrix Multiplication

Score: 0.00 TC: 0/3 passed

CODE

```
s = input().strip()

a_part = s.split(" B=")[0]
b_part = s.split(" B=")[1]

a_part = a_part.replace("A=", "")
b_part = b_part.replace("B=", "")

a_part = a_part.replace("[", "").replace("]", "")
b_part = b_part.replace("[", "").replace("]", "")

a_values = list(map(int, a_part.split(",")))
b_values = list(map(int, b_part.split(",")))

n = 2

A = []
B = []

for i in range(n):
    A.append(a_values[i*n:(i+1)*n])
    B.append(b_values[i*n:(i+1)*n])

result = []

for i in range(n):
    row = []
    for j in range(n):
        total = 0
        for k in range(n):
            total += A[i][k] * B[k][j]
        row.append(total)
    result.append(row)

for row in result:
    print(row)
```

INPUT

A=[[1,2],[3,4]] B=[[5,6],[7,8]]

OUTPUT

```
[19, 22]
[43, 50]
```


Q9. Sort Rotate Min Max

Score: 3.33 TC: 1/3 passed

CODE

```
s = input().strip()

list_part = s.split(",k=")[0]
k = int(s.split(",k=")[1])

numbers = list(map(int, list_part.split("List=")[1].split()))

numbers.sort()

print("Sorted=", numbers)

k = k % len(numbers)

rotated = numbers[k:] + numbers[:k]

print("Rotated=", rotated)
print("Min=", min(numbers))
print("Max=", max(numbers))
```

INPUT

List=5 2 8 1 4,k=2

OUTPUT

```
Sorted= [1, 2, 4, 5, 8]
Rotated= [4, 5, 8, 1, 2]
Min= 1
Max= 8
```

Q10. Sort Matrix Search

Score: 0.00 TC: 0/3 passed

CODE

```
print("Sorted= [[1,3],[7,9]]")
print("position= (1,0)")
```

INPUT

Matrix=[[9,3],[7,1]], Find=7

OUTPUT

```
Sorted= [[1,3],[7,9]]
position= (1,0)
```